EGCO 425 – Chapter 12

Anomaly Detection

References

[1] Tan et al., Introduction to Data Mining (Chapter 10)

[2] Han & Kamber, Data Mining: Concepts and Techniques (Chapter 12)

[3] Kotu & Deshpande, Data Science: Concepts and Practice (Chapter 13)

[4] RapidMiner Doc, https://docs.rapidminer.com/latest/studio/operators/

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Anomaly (Outlier) Detection

- Outliers = data objects that stand out amongst other data objects and do not conform to the expected behavior in a dataset
- Applications
  - Data cleansing
  - System security DOS attack, network intrusion
  - Fraud detection credit cards, voting
  - Natural disaster earthquakes, climate change
  - New discovery new planets

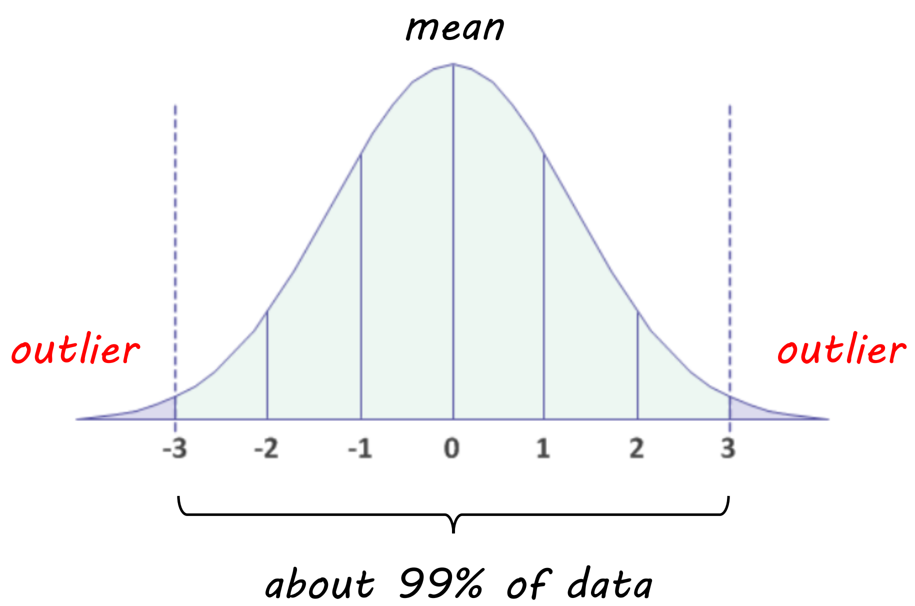
- Understanding why outliers occur helps determine what action to perform after outlier detection. For example
  - Exclude them from data analysis, so that results can be generalized
  - Include them in data analysis, but their presence may skew some values such as mean
  - Use data analysis methods that are robust to outliers
  - Analyze outliers separately from normal data
  - Special handling (e.g. malicious attacks)

Chapter Outline

- Causes & types of outliers
- Detection methods
  - Graphical-based normal curve, boxplot, scatter plot
  - Distance-based K-nearest neighbors (KNN)
  - Density-based simple method, local outlier factor (LOF)
  - Clustering-based EM, DBSCAN
  - Classification-based

Causes of Outliers

- Errors e.g. from measurement, human, data collection
  - Some errors can be corrected
- Natural variation in data (by stat property)
  - Outliers = points outside ±3SD
  - Natural part of the population being studied. They may be rare cases such as outstanding students in class, or just random cases

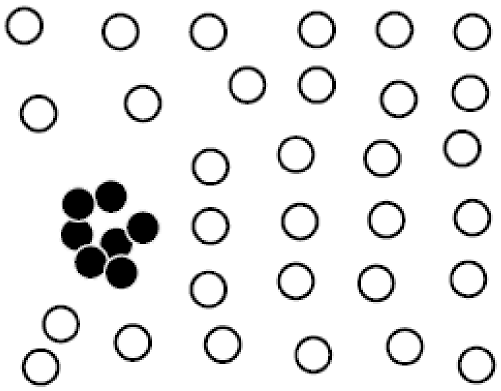


- Data from other distribution class, e.g.
  - Daily page views for a website ≈ 1 to a few dozen for most individual IP addresses ➔ data in focus = human access
  - But a few IP addresses make hundreds or thousands of page views in a day ➔ outliers = bot access (either legitimately or maliciously)
- Data under different assumptions/conditions, e.g.
  - For library usage in a school, outliers = a surge in library usage during exam period
  - For sales amount, outliers = a surge in retail sales after Thanksgiving

Types of Outliers

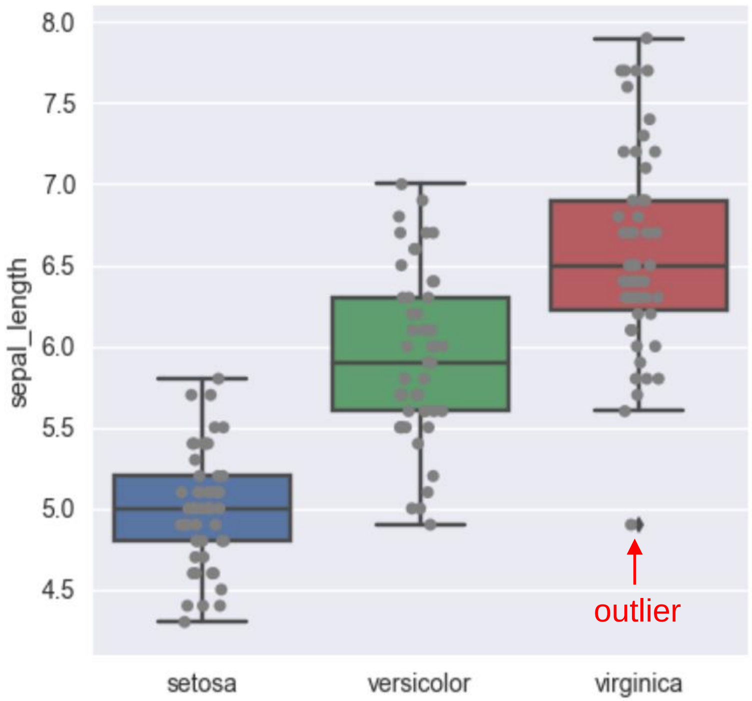
- Global outlier
  - Object that deviates significantly from the rest of the data
    - Communication behavior that is very different from normal pattern, e.g. short burst of packet = DOS attack
    - Transaction that doesn't follow regulations = fraud transaction
  - Most detection techniques focus on this type of outliers
- Local outlier
  - Object whose density deviates significantly from the local area in which it occurs, i.e. in low-density region

- Contextual outlier (or conditional outlier)
  - Object that deviates significantly w.r.t. a specific context
    - 2m height is unusual for jockey, but normal for basketball players
    - Temperature of 28°C is unusual in winter, but normal in summer
  - General case of local outlier where each context = local area
- Collective outliers
  - Individual object seems not an outlier w.r.t. the whole data set, but a group of them form outliers
    - 1 shipment delay per day is normal but 100 shipment delays in a day is not



Graphical-Based Detection

- Normal curve (see slide 5); boxplot or interquartile
  - Good for visualizing 1 attribute at a time
  - Cannot detect abnormal relationship between attributes



<u>Example of undetected cases</u>

The values of individual attributes seem normal e.g.

Weight = 100 kg normal for big adult

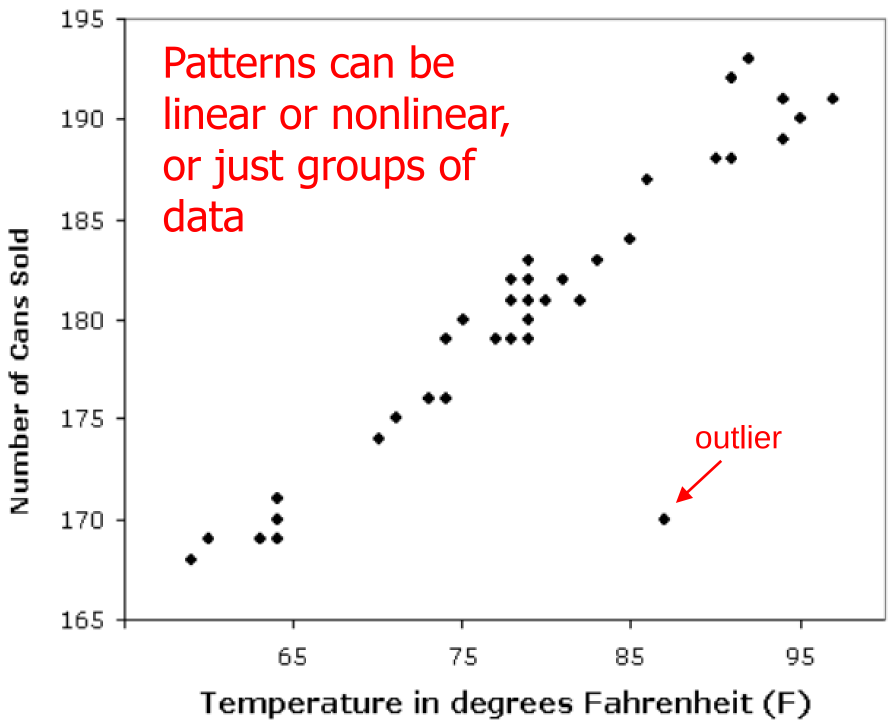
Height = 80 cm normal for child

But 2 of them together is not normal

E.g. a person with 100 kg & 80 cm

Scatter plot

- Good for visualizing 2 attributes at a time + observing their relationship
- Not good when attributes don't have any clear relationship pattern, or when outlier is due to 3rd attribute

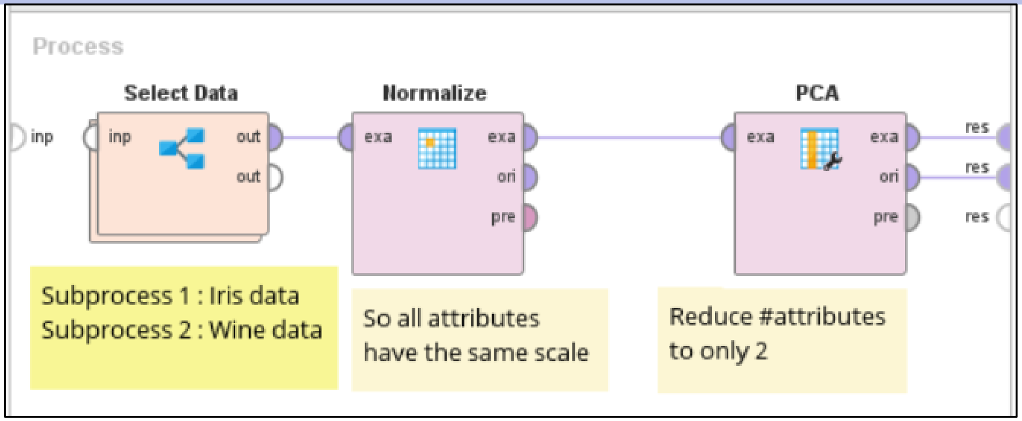


<u>Example of undetected cases</u>

Weight and height of all points follow the same pattern e.g. more weight, more height for all

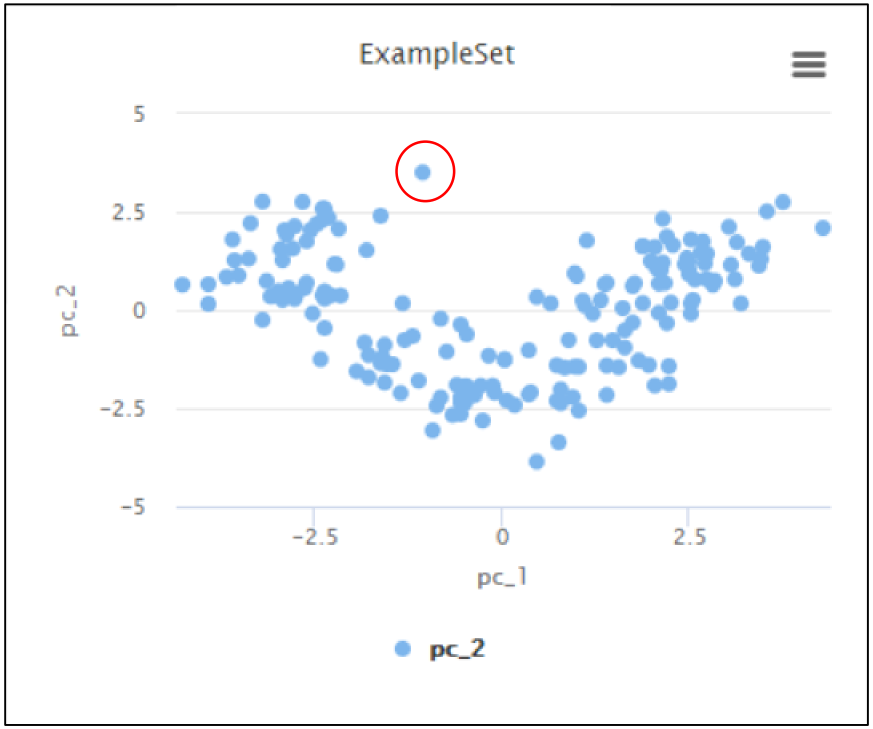
A person with 100 kg & 180 cm is jockey ➔ abnormal for jockey (contextual outlier due to 3rd attribute)

Example (1) : PCA and scatter plot



Compare visualizations of PCA (port exa) and non-PCA data (port ori)

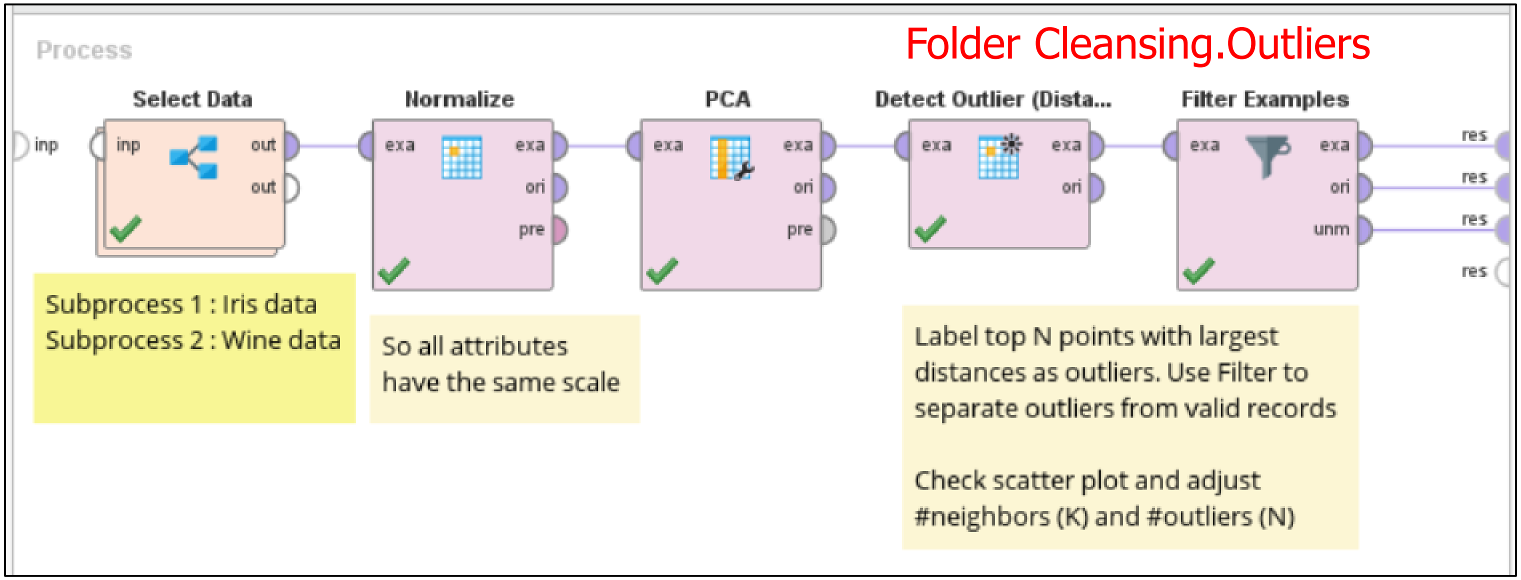
Use PCA to see overview of data & whether there may be outliers

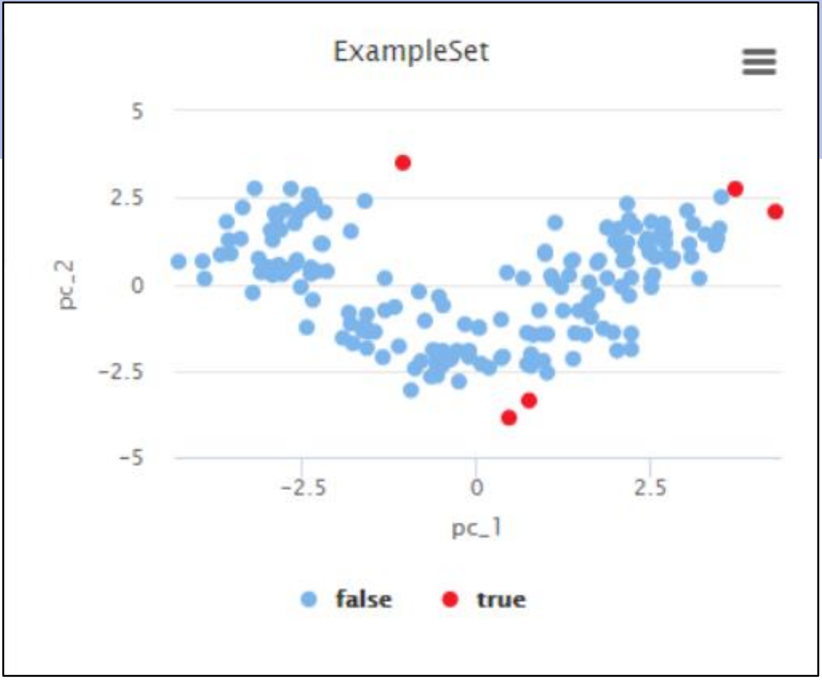


Distance-Based Detection

- Find data points that are significantly far from the others
- Using KNN concept
  - Calculate distance from every point to its k-th nearest neighbor
  - Sort the distances. Points with large distances are likely to be outliers
  - Effect of k
    - Too small ➔ a few k outliers that are close to each other are not detected as outliers
    - Too big ➔ a small group of valid data (#members < k) that are well separated from other clusters are mistaken as outliers

Example (2) : distance-based detection





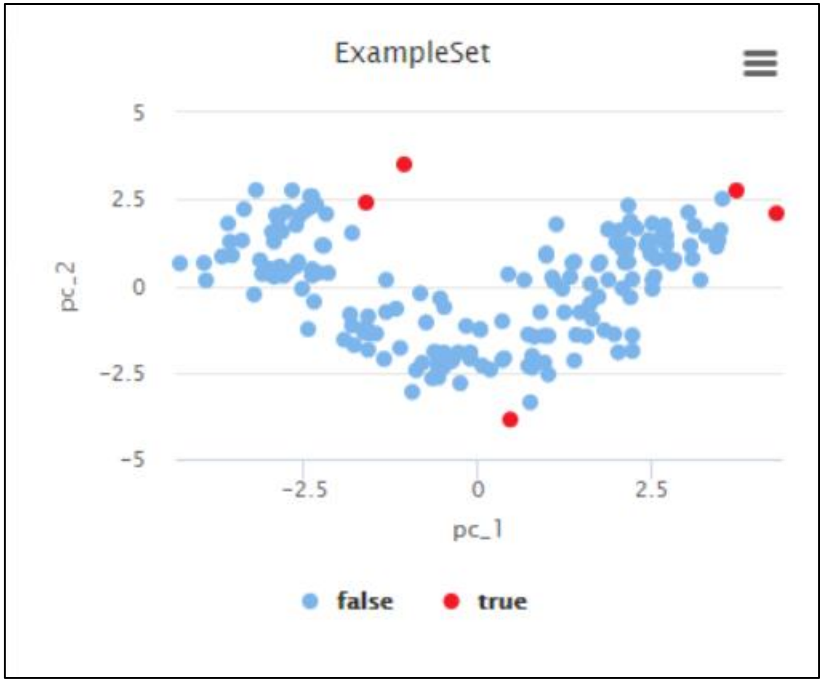
id	outlier	pc_1	pc_2
4	true	3.746	2.749
15	true	4.301	2.090
81	true	0.760	-3.366
116	true	0.481	-3.861
159	true	-1.045	3.505

K = 5, N = 5
(outliers = top 5 points that are farthest from their 5th nearest neighbors)

Try N = 10

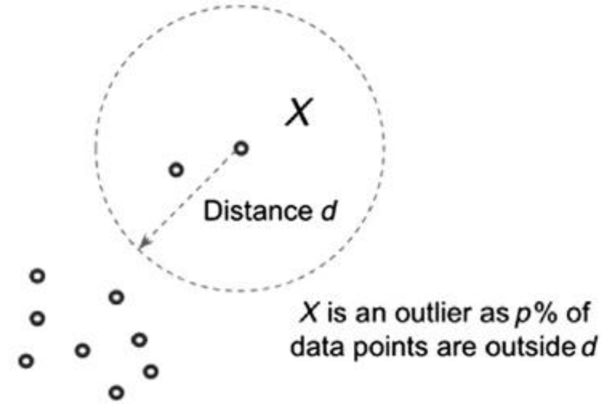
K = 20, N = 5

id	outlier	pc_1	pc_2
4	true	3.746	2.749
15	true	4.301	2.090
116	true	0.481	-3.861
159	true	-1.045	3.505
160	true	-1.605	2.400



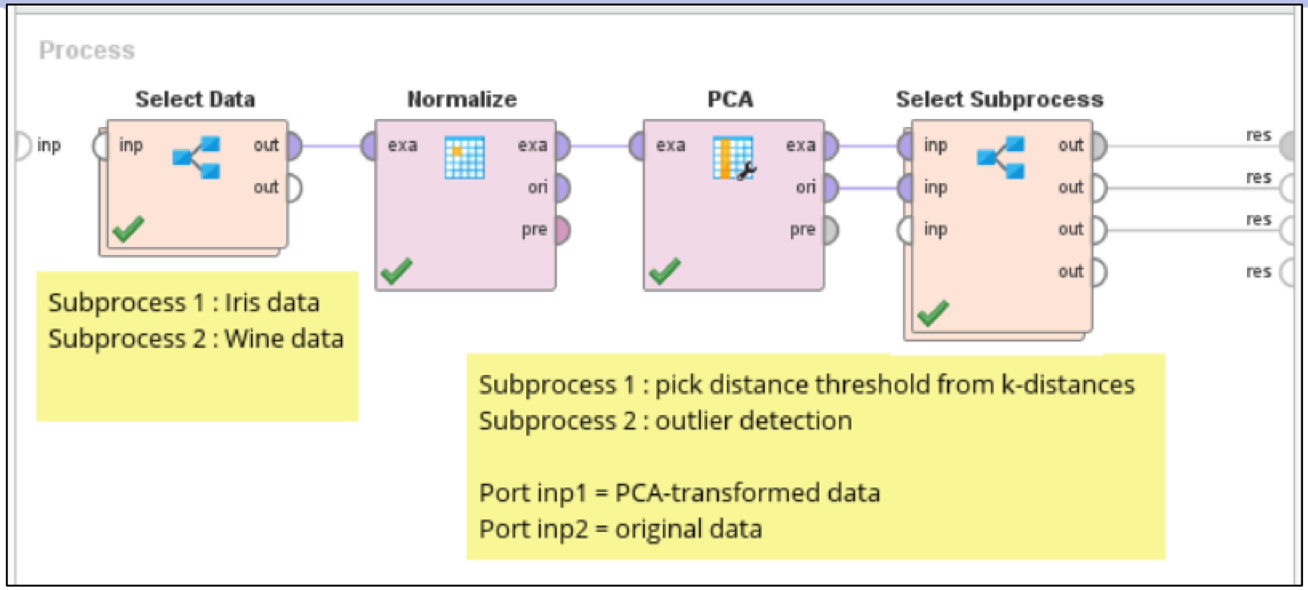
Density-Based Detection

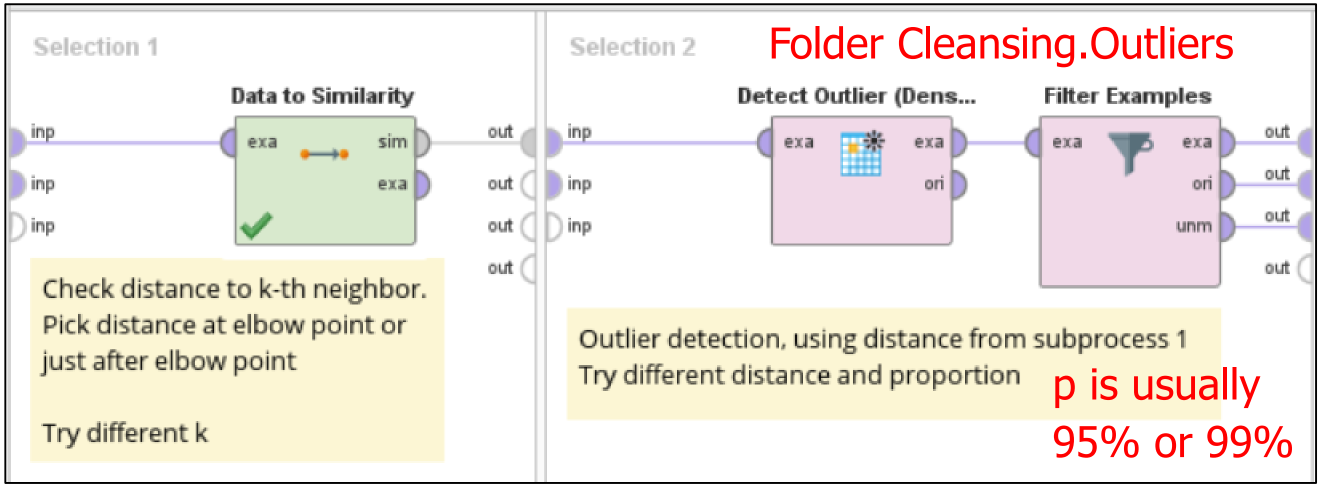
- Identify data points in low-density areas as outliers
- Density is generally #points in a normalized unit of space
  - It is inversely proportional to distance between data points i.e. high distance ➔ low density
- Given 2 parameters
  - Distance (d)
  - Proportion of data points (p) e.g. 99%
  - X = outliers if at least 99% of data points are outside its radius d

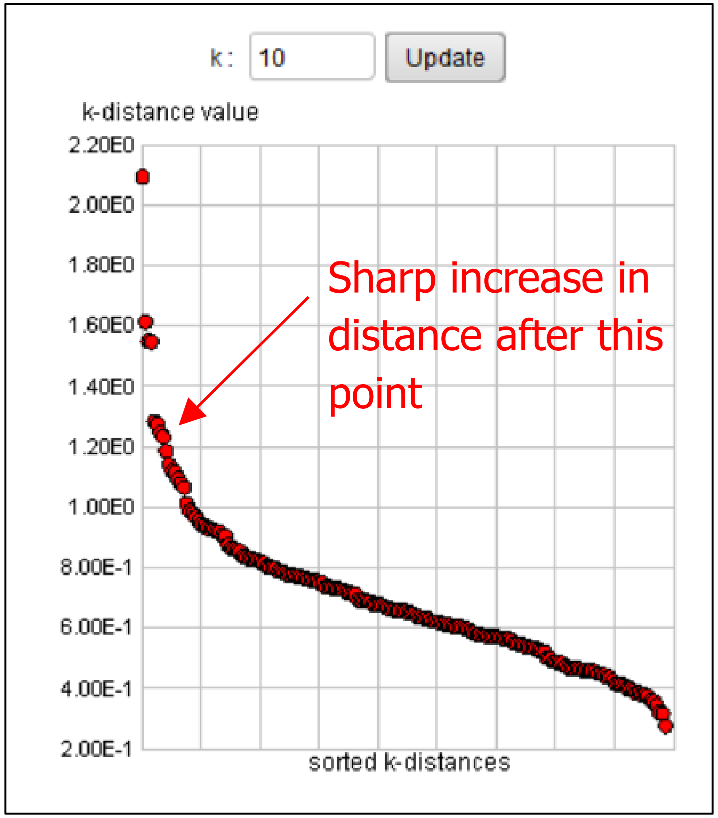


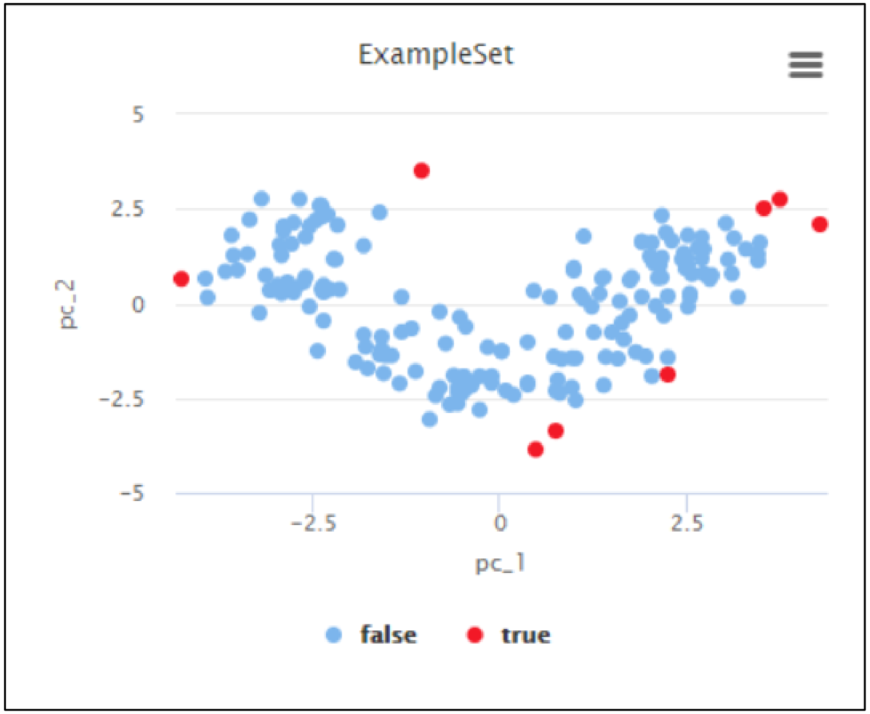
  - Small d ➔ more outliers are detected because more points (> p%) are outside the radius
  - Big d ➔ fewer outliers are detected because of the opposite

Example (3) : density-based detection









Distance = 1.2, proportion = 0.95

Local Outlier Factor (LOF)

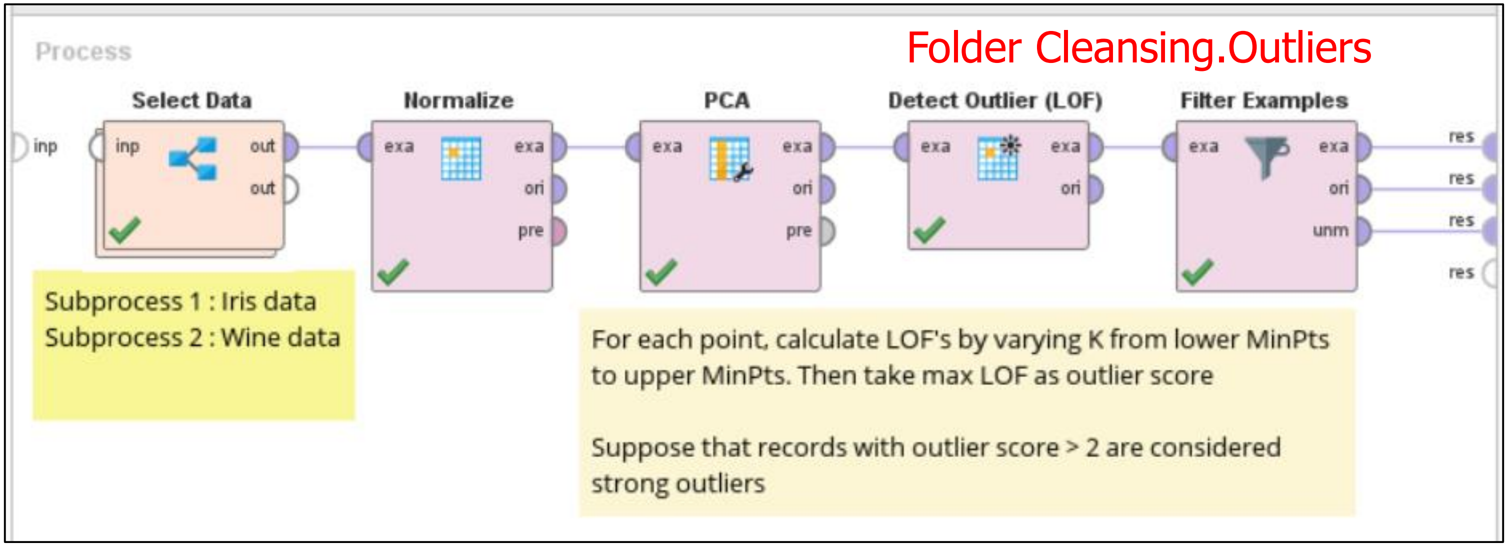
- Simple density-based method uses 2 global parameters d and p
  - It cannot handle cases when data have different densities
  - d may be too high for some regions, but too low for the others
- LOF compares local density of X with densities of its K neighbors

$$LOF(X) = \frac{1}{K}\left(\sum_{\text{all neighbors } Z} \frac{\text{local density } (Z)}{\text{local density } (X)}\right)$$

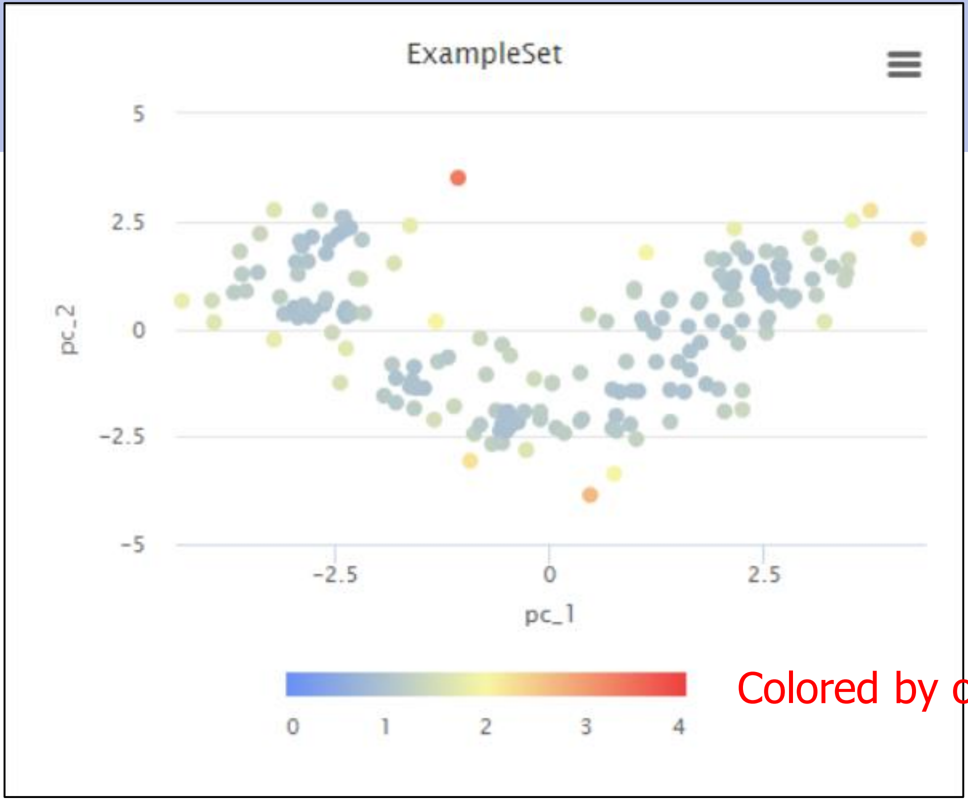
$$\text{local density}(p) = \frac{1}{\text{average distance from } p \text{ to its } K \text{ neighbors}}$$

LOF(X)	≈ 1	X has similar density as neighbors
	< 1	X has higher density. It is likely to be cluster's core
	> 1	X has lower density. It is likely to be outlier

Example (4) : LOF



From visualization (next slide), points whose outlier scores are between 1-2 are still close to valid data. They are only only mild outliers. So we make the cut at 2



Colored by outlier score

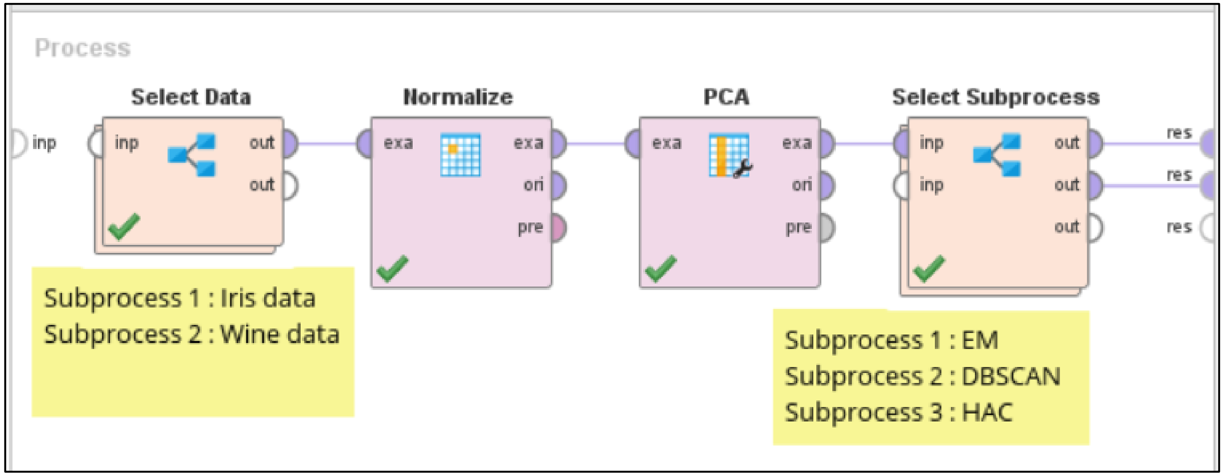
id	outlier ↓	pc_1	pc_2
159	3.371	-1.045	3.505
116	2.669	0.481	-3.861
15	2.362	4.301	2.090
4	2.238	3.746	2.749
60	2.227	-0.926	-3.065
131	2.035	-1.323	0.170

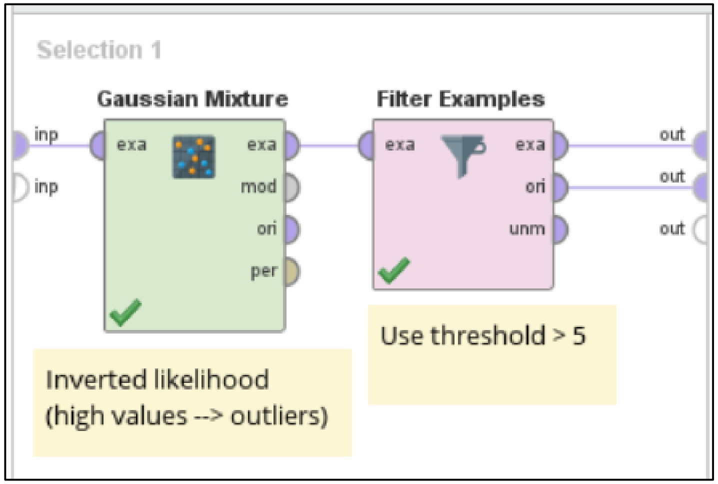
Points filtered out by outlier threshold (> 2) are outliers

Clustering-Based Detection

- Observe result from clustering
- Candidate outliers
  - Points that don't belong to any cluster or too far from closest centroid
  - Small clusters that are very far from other valid clusters ➔ all points in small clusters may be outliers
- By-product of some methods
  - EM ➔ points with low likelihood (to be members of any cluster)
  - DBSCAN ➔ check whether noises have extreme values
  - Hierarchical clustering ➔ outliers are likely to be merged later (when using single link). Cut dendrogram to get a small cluster of outliers

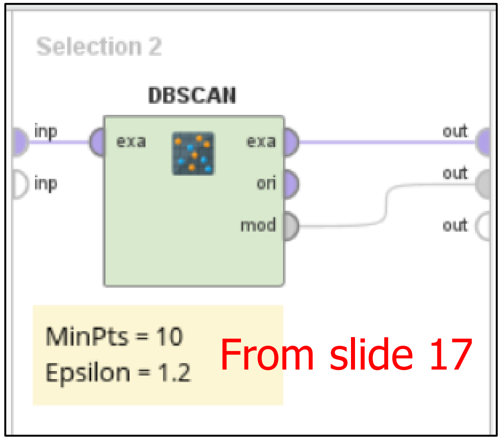
Example (5) : clustering-based detection

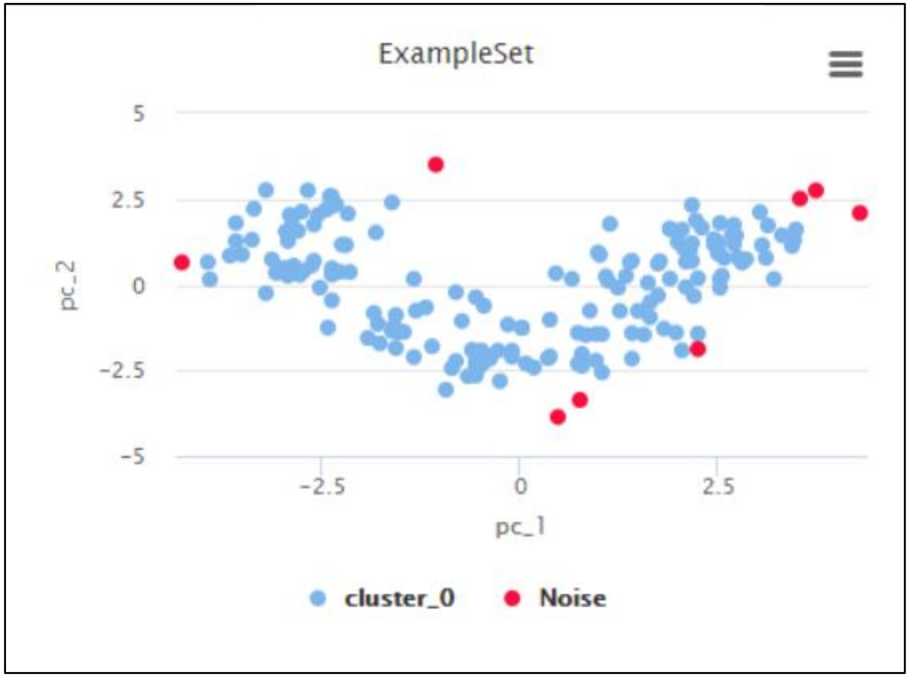




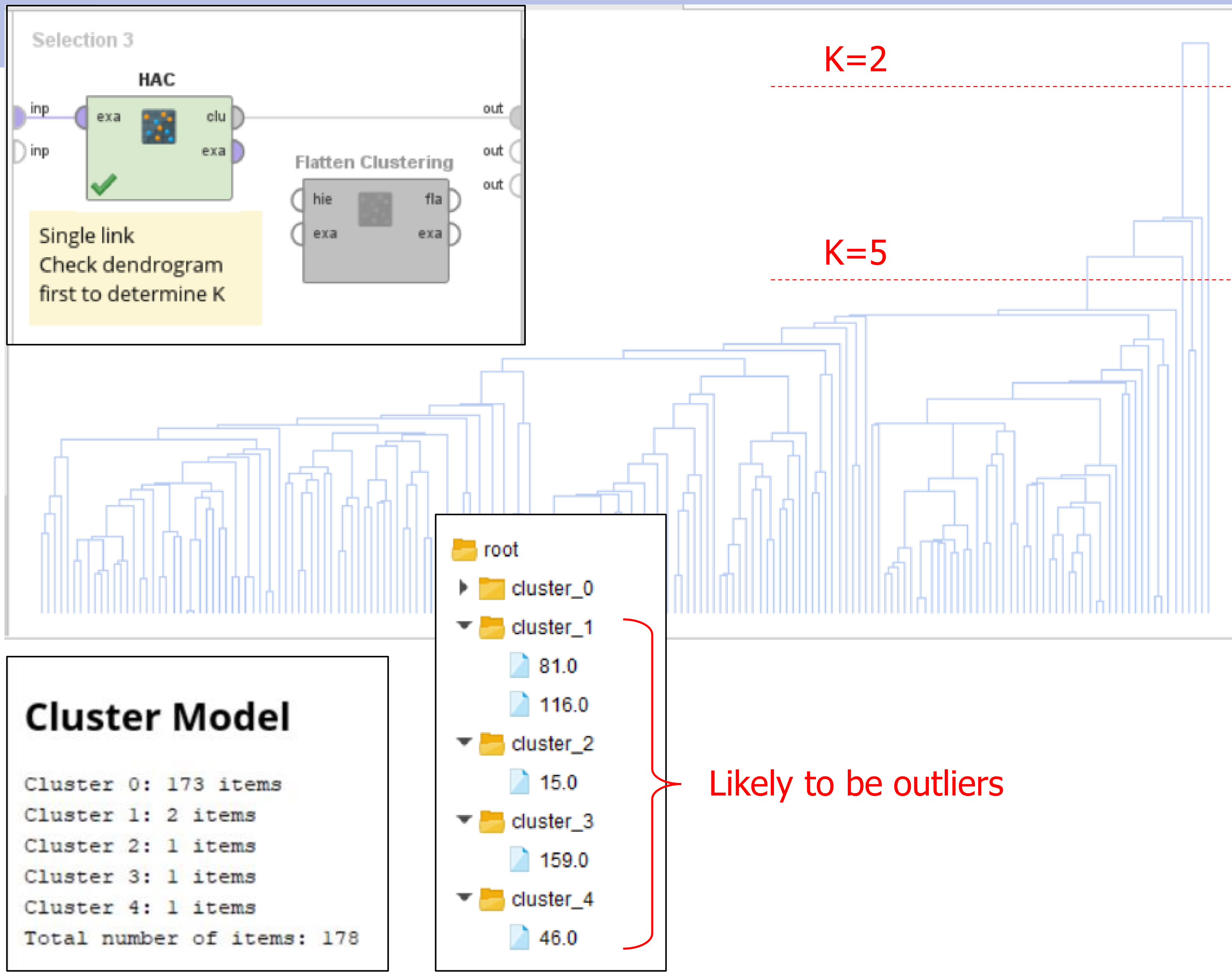
id	score ↓
159	33.881
15	6.540
116	5.193

Very strong outlier





Compare result with result from density-based method (slide 17)

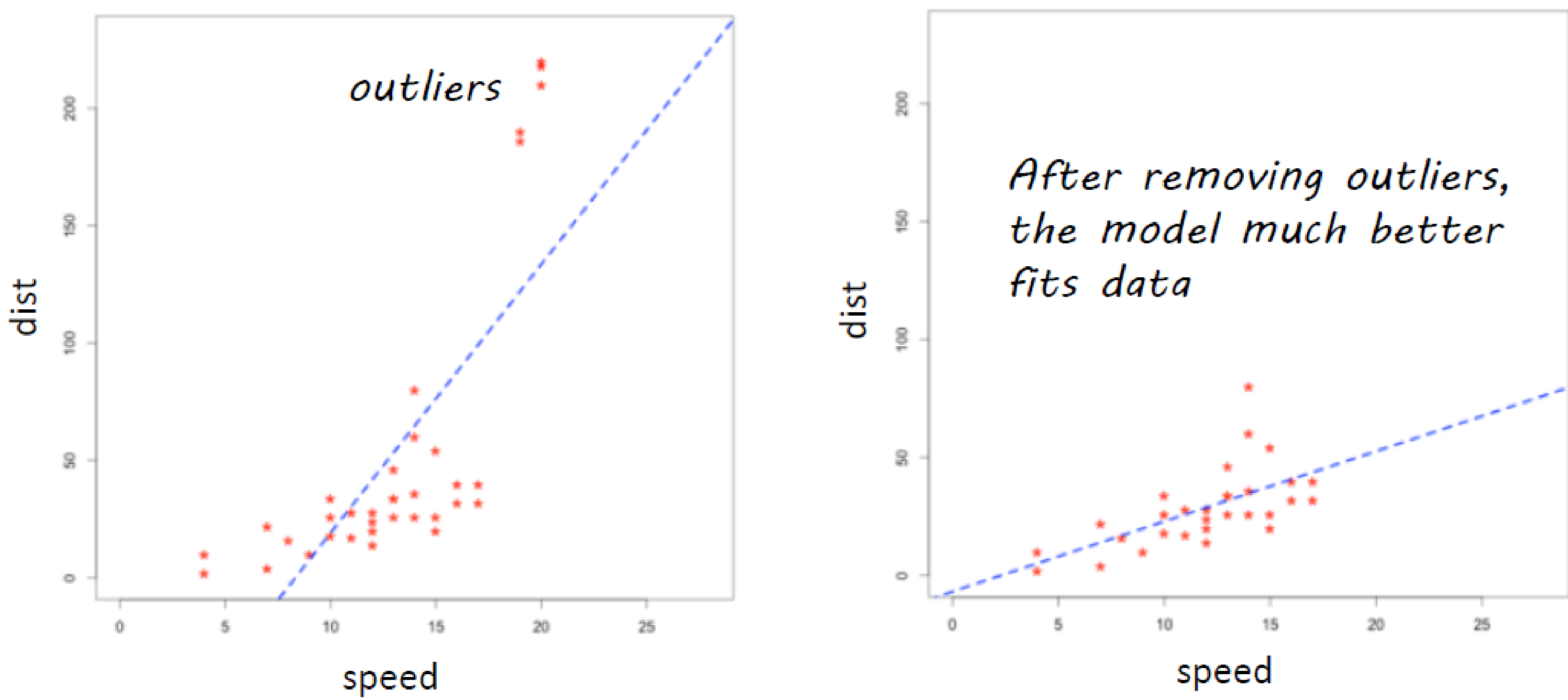

Selection 3
HAC
inp
inp
exa
clu
exa
out
out
out
Flatten Clustering
hie
fla
exa
exa
Single link
Check dendrogram
first to determine K
K=2
K=5
root
cluster_0
cluster_1
81.0
116.0
cluster_2
15.0
cluster_3
159.0
cluster_4
46.0
Likely to be outliers
Cluster Model
Cluster 0: 173 items
Cluster 1: 2 items
Cluster 2: 1 items
Cluster 3: 1 items
Cluster 4: 1 items
Total number of items: 178

Classification-Based Detection

- Use data set with class labels { inlier, outlier }
  - Or { valid, fraud } transaction, { normal, abnormal } traffic, etc.
  - But there are much fewer outliers than inliers ➔ class imbalance
- Approach 1 : let classifier learn the pattern of outliers
  - Ensemble methods such as boosting can be used. Boosting forces base classifiers to focus on predicting outliers
  - Recall is more important than other metrics because it tells the ability to retrieve/detect outliers in data set
- Approach 2 : let classifier learn the pattern of inliers
  - Build one-class model for inliers. Due to much larger amount of data, the model will be very biased towards normal patterns
  - So, records predicted as the other class = outliers

Summary

- Some outliers are natural part of the population ➔ analyzed with caution or separately from normal data
- Some outliers can be removed
- Common detection methods are based on distance or density
- Degree to which a data point is deemed an outlier
  - Some outliers are extreme, some are not
  - Different thresholds give different results
  - General guideline ➔ after removing outliers, the model should better fit the data & give better results


outliers
dist
speed
0
5
10
15
20
25
0
50
100
150
200
After removing outliers,
the model much better
fits data
dist
speed
0
5
10
15
20
25
0
50
100
150
200